Pathfinder

Software Requirements Specification

Overview

Project Name	Pathfinder
Team	Patrick King- Project Manager Maddie McDowell- Front-End and UI Design Nic Ortiz- Front-End Thomas Powell- Back-End
Brief Description	Pathfinder is an all-in-one career development web application designed to guide users through every stage of their professional and educational journey
Project Scope	This SRS describes the software requirements for "Pathfinder," a web application developed using React, HTML, CSS, and Node.js. It is designed for high school and college students entering the workforce, offering comprehensive assistance through a personalized user interface. The core features include career guidance, job matching, resume analysis, and an AI-powered chatbot assistant.

Component Breakdown

Front-End	• **Landing Page:** Login/Sign-Up functionality with responsive design for various devices. • **Dashboard:** Displays personalized user information and provides access to core features. • **Chatbot UI:** AI chatbot accessible on all pages for quick assistance. • **Profile Setup Page:** Allows users to enter and edit personal and career preference details.
Back-End	• **Landing Page:** Login/Sign-Up functionality with responsive design for various devices. • **Dashboard:** Displays personalized user information and provides access to core features. • **Chatbot UI:** AI chatbot accessible on all pages for quick assistance. • **Profile Setup Page:** Allows users to enter and edit personal and career preference details.
AI-Powered Chatbot	• **Query Processing:** Routes user queries to OpenAI API for response generation. • **Context Management:** Retains session context for a conversational experience.
Database and Data Storage	• **Firebase Database:** Stores user profiles, preferences, and historical data securely.
Deployment	• Hosted on cloud platforms (e.g., AWS or Google Cloud) for scalability and reliability.

Requirements

Required States and Modes	• **Landing Page State:** Initial page for logging in or signing up. • **Dashboard State:** Centralized hub displaying user-specific tools and information. • **Settings State:** Allows customization of account settings and preferences.
Functional Requirements	• **FR1:** The website shall support user authentication via Firebase. • **FR2:** The website shall provide personalized career suggestions based on a questionnaire. • **FR3-1:** The website shall match jobs based on user-selected career paths. • **FR3-2:** The job-matching system shall refresh results every 24 hours. • **FR4:** The website shall analyze uploaded resumes and suggest improvements in specific areas. • **FR5-1:** The website shall include an AI chatbot assistant for career-related guidance. • **FR5-2:** The chatbot shall be accessible through a floating button in the bottom-right corner of all pages. • **FR5-3:** The chatbot shall interact with users via a text-based interface with an optional voice input feature. • **FR5-4:** Responses from the chatbot shall be generated via OpenAI API, with fallback responses for common queries. • **FR5-5:** The chatbot shall offer direct links to additional resources for further guidance. • **FR6:** Users shall be able to update career preferences and settings in their profile. • **FR7:** The dashboard shall notify users of new job matches and updates to their career paths or resumes.
External Interface Requirements	• **EIR1:** Integrate with job board APIs for fetching relevant job listings. • **EIR2:** Use Firebase for secure data storage and authentication.
Internal Data Requirements	• **IDR1:** User data, including profile and preferences, shall be securely stored in a Firebase database. • **IDR2:** Questionnaire results and career path data shall be retrievable for real-time user interactions.
Adaptation Requirements	• **AR1:** The website shall adapt seamlessly to varying screen sizes, ensuring usability on mobile, tablet, and desktop devices.
Security Requirements	• **SecR1:** User data, such as email and passwords, shall be encrypted

	during transmission and storage. • **SecR2:** Only authenticated users shall have access to personalized features.
Environmental Requirements	• **ER1:** The application shall support major browsers, including Chrome, Firefox, Safari, and Edge.
Computer Resource Requirements	• **CRR1:** The system shall perform optimally on low-spec mobile devices.
Software Quality Factors	• **SQF1:** The interface shall be intuitive, minimizing the need for external user guidance. • **SQF2:** Downtime for maintenance shall be communicated in advance to users.
Design and Implementation Constraints	• **DIC1:** The development process shall be constrained by the capabilities of React, Node.js, Firebase, and OpenAI APIs.
Personnel-related Requirements	• **PRR1:** The project team must include at least one full-stack developer proficient in the chosen tech stack.
Training-related Requirements	• **TRR1:** A built-in tutorial shall guide users through the application's features during the initial setup.
Other Requirements	• **OR1:** Accessibility features such as screen reader compatibility and keyboard navigation shall be implemented.

Miscellaneous

Precedence and Criticality of Requirements	• Security and usability requirements are critical and shall take precedence in the development lifecycle.
Qualification Provisions	• **Usability Testing:** Evaluate user interface intuitiveness. • **Performance Testing:** Assess response times and scalability. • **Security Testing:** Verify data protection measures against breaches.
Requirements Traceability	• A requirements traceability matrix shall map each requirement to its corresponding implementation and test case.
Notes	• This document is subject to revisions based on iterative feedback from stakeholders and users.
Appendices	• **A. Glossary:** Contains technical terms used in this document. • **B. Revision History:** Logs updates to this SRS.
Evaluation Criteria	The project's success will be determined by: • Adherence to the specified requirements. • Feedback from beta testers on usability and functionality. • System performance under expected user loads.
Acronyms and Abbreviations	• **SRS:** Software Requirements Specification • **UI:** User Interface • **API:** Application Programming Interface